v1.0Boson Deep Dive_ A Universe

V1.0 to V2.0 Versioning Delta — Tardigradia SubParticles Engine

Architecture: One-Gauge-Boson Universe Architecture **Type:** Scientific Change Log | **Date:** June 2026 **Applies to:** `v1.0Boson Deep Dive_ A Universe V2.0.md`

Revision Necessity — Four Drivers

1. **Experimental discoveries 2022-2026** — LHC Run 3, LIGO O4, KATRIN 2024, NIF ignition, and other landmark results supersede V1.0 physics parameters
2. **New confirmed particles and phenomena** — Pines' Demon (2023), HH production evidence (2024), IceCube tau neutrino (2023), and others add entirely new simulation modes
3. **Platform upgrade** — WebGPU (Chrome 113+ 2023) replaces WebGL; enables 10^7+ particles
4. **Precision improvements** — CODATA 2022 constants, FLAG 2024 lattice QCD, NuFIT 5.3 oscillations

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V1.0 vs V2.0 Specific Changes

Parameter	V1.0	V2.0	Severity	Consequence if Not Updated
W boson mass	M_W = 80.377 GeV (SM)	CDF 2022 (7σ anomaly) vs CMS 2024 (SM-consistent). World average preserved at 80.377 GeV; tension flag added	HIGH	CDF anomaly unresolved; simulation should reflect tension state
Gluon OAM	Not in V1.0	sPHENIX 2024: gluon OAM L_g~0.2*hbar confirmed. Updates proton spin sum. New data field: gluon_OAM = 0.2	MEDIUM	Gluon helicity data structure incomplete without OAM term
Graviton mass from LIGO	Pre-O4 bound	LIGO O4 (2024): m_g < 1.27×10^-23 eV. V2.0 graviton propagator updated	LOW	More precise bound improves graviton dispersion simulation
Photon mass from FRB	Bound < 10^-18 eV	Wu et al. Nature Astronomy 2024: m_γ < 7.1×10^-23 eV from FRB dispersion	LOW	13× tighter mass constraint improves photon propagator accuracy

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Simulation Impact Summary

The changes above drive the following modifications to the game engine architecture:

- **New particle types:** W
- **Parameter updates:** All values in `static constexpr` declarations refreshed from 2022-2026 data
- **WebGPU migration:** Particle update loop moves from CPU JavaScript to WGSL compute shaders
- **New interaction modes:** Triggered by new experimental discoveries

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Unchanged Architecture

All foundational philosophy is **carried forward unchanged**: - Worldline Monism (One-Particle Universe) ontology - Proper time τ as primary particle identifier - Intrinsic vs. Extrinsic property distinction - Benevolence metric (C·F·S product formula) - Object-Oriented data structure (struct ParticleInstance extends to V2.0)

End of Versioning Delta | v1.0Boson Deep Dive_ A Universe V1.0 archived | V2.0 is the active specification as of June 2026